

where for the last equality, “ $\supseteq$ ” is clear, while for “ $\subseteq$ ” we have to work. In fact, for this we can choose some $\mathbf{x} \in \text{relint}(F)$, which satisfies $A'\mathbf{x} = \mathbf{1}$ and $A''\mathbf{x} < \mathbf{1}$, and rewrite $\mathbf{c}A = \mathbf{c}'A' + \mathbf{c}''A''$. Then by

$$1 = (\mathbf{c}A)\mathbf{x} = (\mathbf{c}'A' + \mathbf{c}''A'')\mathbf{x} = \mathbf{c}'(A'\mathbf{x}) + \mathbf{c}''(A''\mathbf{x}) \leq \mathbf{c}'\mathbf{1} + \mathbf{c}''\mathbf{1} = \mathbf{c}\mathbf{1} = 1$$

we have $\mathbf{c}''(A''\mathbf{x}) = \mathbf{c}''\mathbf{1}$, which by $A''\mathbf{x} < \mathbf{1}$ implies $\mathbf{c}'' = \mathbf{0}$. $\square$

Corollary 2.13. *Let P be a polytope with $\mathbf{0} \in \text{int}(P)$, and let $F, G \in L(P)$ be faces of P . Then*

- (i) $F^\diamond$ is a face of P^Δ ,
- (ii) $F^{\diamond\diamond} = F$, and
- (iii) $F \subseteq G$ holds if and only if $F^\diamond \supseteq G^\diamond$.

Corollary 2.14. *The face lattice of P^Δ is the opposite of the face lattice of P :*

$$L(P^\Delta) \cong L(P)^{op}.$$

This, in particular, completes the proof of Theorem 2.7, the last two parts of which we had deferred (remember?). It means that for every statement about the combinatorial structure of polytopes, there is a “polar statement,” where the translation reverses inclusion of faces, and interchanges

$$\begin{array}{ccc} \emptyset = \hat{\mathbf{0}} & \longleftrightarrow & \hat{\mathbf{1}} = P \\ \text{vertices} & \longleftrightarrow & \text{facets} \\ \text{edges} & \longleftrightarrow & \text{ridges} \\ \dots & \longleftrightarrow & \dots, \quad \text{etc.} \end{array}$$

Note that polarity also identifies the face lattices of facets with (the opposites of) the face lattices of vertex figures. Finally, it says that the “polar” combinatorial descriptions of polytopes, as $\mathcal{V}$ -polytopes in terms of vertices, and as $\mathcal{H}$ -polytopes in terms of facets, are logically equivalent.

Nevertheless, the metric properties of the polarity construction depend on the location of $\mathbf{0}$ in P , whereas the combinatorial ones do not. This motivates to define that two polytopes P and Q are *combinatorially polar* if $L(P) \cong L(Q)^{op}$. Thus the construction of P^Δ establishes the existence of a combinatorially polar polytope for every polytope P .

2.4 The Representation Theorem for Polytopes

This section has a (by now) simple task: to state and prove the general representation theorem for polytopes. There is no real work left to do: we have assembled all the ingredients, notably Fourier-Motzkin elimination, the Farkas lemmas, Carathéodory’s theorem, and polarity. One new term appears in its statement: the *k-skeleton* of a polytope is the union of its k -dimensional faces.

Theorem 2.15 (Representation theorem for polytopes).

A subset $P \subseteq \mathbb{R}^d$ is a polytope if and only if it can be described in any of the following (equivalent) ways:

- (1) an affine projection of a simplex,
- (2) all the convex combinations of a finite point set,
- (3) all the convex combinations of the vertex set $\text{vert}(P)$,
- (4) the union of all simplices spanned by a finite set of points,
- (5) the projection of the d -skeleton of a simplex,
- (6) a bounded intersection of (closed) halfspaces,
- (7) a bounded intersection of facet-defining (closed) halfspaces, one for each facet, and of the affine hull of P .

Proof. The equivalence of (1) and (2) is from the definitions of a convex hull and a simplex. It is just an example of translation of a geometric statement (1) into an algebraic one (2). Similarly, (4) $\iff$ (5) is such a translation.

The equivalence of (2) and (3) is from Proposition 2.2, while the equivalence of (1) and (5) is from Carathéodory's theorem 1.15(ii).

The equivalence of (2) and (6) is the main theorem on polytopes, Theorem 1.1, which we proved by Fourier-Motzkin elimination. Instead, we could also argue that (2) $\iff$ (6) follows from polarity.

Finally, for (3) $\iff$ (7) we reduce this to the full-dimensional case, and then use that the facets of P^Δ correspond to the vertices of P under polarity, by Theorem 2.12 and its corollaries. In particular, the facet-defining inequalities are uniquely determined (if we write them as $\mathbf{a}_i \mathbf{x} \leq 1$), and none of them can be deleted. $\square$

2.5 Simplicial and Simple Polytopes

Proposition 2.16. For any d -dimensional polytope P , the following conditions are equivalent:

- (i) every facet of P is a simplex, i.e., P is *simplicial*,
- (ii) every proper face of P is a simplex,
- (iii) every facet has d vertices,
- (iv) every k -face has $k + 1$ vertices, for $k \leq d - 1$,
- (v) every lower interval $[\hat{0}, F] \subseteq L(P)$ in the face lattice with $F \neq \hat{1}$ is a boolean poset.